FEATURES OF FIXED-INCOME SECURITIES

Learning Outcome

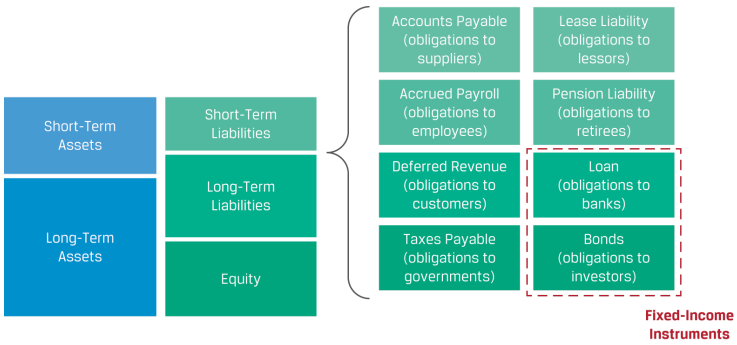
- describe the features of a fixed-income security

Fixed-income instruments are debt instruments, such as loans and bonds. **Loans** are debt instruments formed and governed by a private agreement usually between an individual or company and a financial intermediary, such as a bank. **Bonds** or **fixed-income securities** are more standardized contractual agreements between larger issuers and investors. A bond issuer borrows money most often to fund operations or capital expenditures. Bond investors are lenders who provide funds to the issuer in exchange for interest payments and future repayment of principal. While corporate issuers tend to have, at most, one or two types of equity securities outstanding, they often have many types of debt obligations outstanding, each with distinct features, such as time to maturity, seniority, and currency.

Which Liabilities Are Fixed-Income Instruments?

Earlier modules in corporate issuers and financial statement analysis discussed the balance sheets of corporate issuers composed of assets and the liabilities and equity that finance them. Liabilities are broadly defined by accounting standards as present obligations to transfer economic resources as a result of past events. This definition encompasses many types of obligations, including amounts that an issuer owes to suppliers, customers, employees, governments, retirees, lessors, and so on.

Not all liabilities are fixed-income instruments (or "debt"), but all fixed-income instruments are liabilities. In these modules on fixed income, from the perspective of a corporate issuer, we are focused only on loans and bonds: instruments that can be settled in cash and for which the counterparty is an investor or a bank. Other liabilities, particularly leases and pension obligations, share some characteristics with fixed-income instruments but are outside the scope of these modules.



Note that government issuers tend to be financed by bonds, not by loans, though some exceptions exist—for example, loans from supranational organizations such as the International Monetary Fund (IMF).

The committed periodic cash flows of a bond distinguish it from equity securities. Exhibit 1 shows the cash flows of a fixed-rate bond issued by Bright Wheels Automotive (BRWA) Corporation.

Exhibit 1:

BRWA Corporation Bond Cash Flows

Rate Your Confidence
High
Medium
Low
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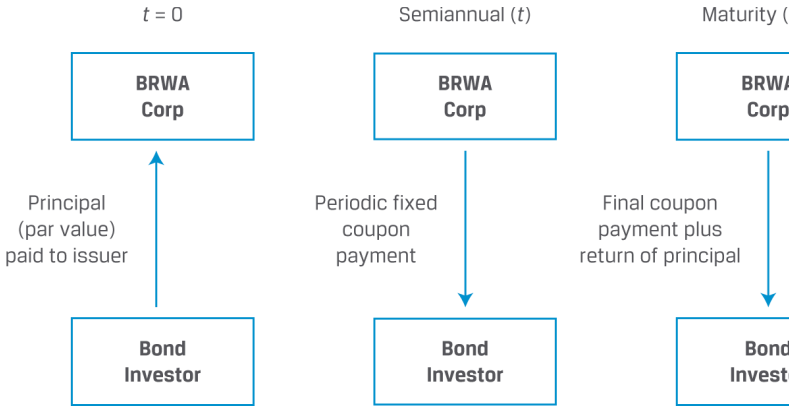
Category
Fixed-Income Instrument Features

Related Questions:
Practice questions related to this topic



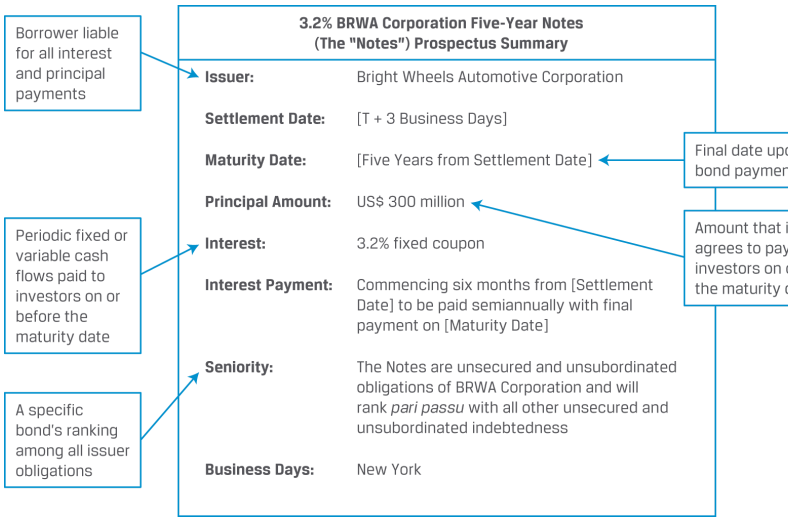
▶ Expand / collapse Extended Description

At issuance, investors purchased the BRWA bond in exchange for cash equal to the bond **principal** or the amount borrowed, which is also referred to as the bond's **par value**. On each interest payment date, BRWA commits to pay bond investors an interest or **coupon** payment equal to the coupon rate times the par value of the bond. For periods shorter than a year, the annual coupon amount is divided into smaller equal periodic payments. On the bond's maturity date, BRWA pays the final fixed coupon and the principal amount to investors.

The key features of each bond issue are usually summarized in a prospectus. Exhibit 2 outlines the terms of BRWA Corporation's bond issue.

Exhibit 2:

BRWA Corporation Bond: Brief Summary of Terms



3.2% BRWA Corporation Five-Year Notes (The "Notes") Prospectus Summary	
Issuer:	Bright Wheels Automotive Corporation
Settlement Date:	[T + 3 Business Days]
Maturity Date:	[Five Years from Settlement Date]
Principal Amount:	US$ 300 million
Interest:	3.2% fixed coupon
Interest Payment:	Commencing six months from [Settlement Date] to be paid semiannually with final payment on [Maturity Date]
Seniority:	The Notes are unsecured and unsubordinated obligations of BRWA Corporation and will rank *pari passu* with all other unsecured and unsubordinated indebtedness
Business Days:	New York

▶ Expand / collapse Extended Description

Key bond features include the issuer, time to maturity, principal amount, coupon rate and frequency, seniority, and contingency provisions.

Issuer

A bond issuer can be any legal entity and is liable for all interest and principal payments. Government sector issuers include national (also termed sovereign) or local governments, supranational organizations (such as the World Bank), and quasi-government entities (agencies owned or sponsored by governments, such as postal services or national railways). Because sovereign bonds are backed by the taxation and fiscal power of the issuing government, they usually represent the lowest-credit-risk bond in each region. Private sector issuers include corporate issuers and special purpose entities created to

Maturity

A bond's **maturity** is the date of the final payment the issuer makes to investors, and the **tenor** refers to the remaining time to maturity. Fixed-income securities with a tenor one year or less at issuance are known as **money market securities**, examples of which include government Treasury bills and commercial paper issued by corporations. Bonds with tenors longer than one year at issuance are called **capital market securities**. **Perpetual bonds** are a less common bond type with no stated maturity. Public sector entities were the first issuers of perpetual bonds, and current examples include local governments and local authorities, as well as certain bonds issued by banks to meet regulatory capital requirements. Perpetual bonds are still distinct from equities, however, in that they have contractually defined cash flows, no voting rights, and greater seniority in the capital structure.

Airport Authority Hong Kong's Perpetual Bonds

Airport Authority Hong Kong (AAHK) is a statutory body responsible for operating and developing Hong Kong International Airport, one of the world's busiest cargo airports and a major passenger hub connecting to over 200 global destinations. In late 2020, following a sharp decline in passenger volume during the COVID-19 pandemic, AAHK announced a two-part, USD1.5 billion perpetual bond issue to fund construction of a third runway and for general corporate purposes. Asian and European investors demonstrated confidence in the airport's recovery of passenger volume, placing orders totaling more than 10× the offer amount.

Principal (Par or Face Value)

The principal, par value, or **face value** is the amount an issuer agrees to repay to investors at maturity. In the BRWA Corporation example, the principal amount of USD300 million is repaid at maturity, which is five years from issuance. Certain instruments may repay the principal in equal or variable increments over time. A common example is a mortgage loan, for which borrowers make a monthly payment composed of both principal repayment and interest. The principal is repaid over time and not in a lump sum at maturity.

Coupon Rate and Frequency

A bond's interest can be paid as

- a fixed coupon paid on specified dates,
- a variable coupon determined and paid on specified dates, or
- part of a single payment with the principal at maturity.

Fixed-coupon bond payments usually involve uniform payments at monthly, quarterly, semi-annual, or annual intervals. Corporate bonds tend to pay semiannually.

Bonds with variable interest payments are called **floating-rate notes** (FRNs). An FRN coupon is determined as a combination of a **market reference rate** (MRR) and an issuer-specific spread referred to as the **credit spread**. The MRR is a standard borrowing or lending rate for issuers with the highest creditworthiness or lowest default risk for different currencies and maturities. MRRs were historically determined by a poll of lenders (Libor) but transitioned to an average of observed market transaction rates.

The credit spread is set at the time of FRN issuance, is usually constant over the bond's life, and is expressed in basis points (bps), or hundredths of a percentage point. The higher an issuer's credit quality, the lower the spread. In contrast, the MRR resets periodically throughout the bond's life based on market factors. As the MRR changes, the FRN coupon rate and interest payment change accordingly.

bonds. Zero-coupon bonds are typically issued at a discount to par; the difference between the issuance price and par value represents a cumulative interest payment at maturity.

Seniority

A debt issue's **seniority** or priority of repayment among all issuer obligations is an important determinant of risk. **Senior debt** has priority over other debt claims in the case of bankruptcy or liquidation. **Junior debt**, or **subordinated debt**, claims have a lower priority than senior debt and are paid only once senior claims are satisfied. The BRWA Corporation bonds are senior debt and would be repaid before any BRWA subordinated debt in the event of bankruptcy.

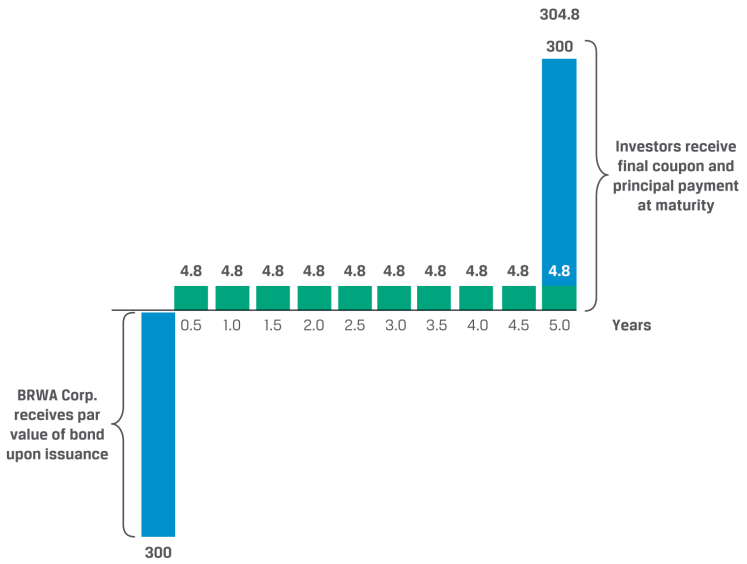
Contingency Provisions

A **contingency provision** is a clause in a legal agreement that allows for an action if an event or circumstance occurs. The most common contingency provision for bonds involves **embedded options**—specifically, call, put, and conversion to equity options. These resemble option contracts but cannot be traded separately from the bond itself. However, the value of these embedded options may be established by comparing the value of a bond with a contingency provision with that of an otherwise similar standard bond from the same issuer. The BRWA bond does not contain any contingency provisions, but this feature will be explored in depth in later lessons.

We can use the features of a bond issue to model its cash flows. Exhibit 3 illustrates the cash flows over the life of the five-year, fixed-coupon BRWA Corporation bond described earlier.

Exhibit 3:

BRWA Corporation Bond Cash Flows



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At the time of issuance, investors purchase the bond and pay cash to BRWA Corporation equal to the USD300 million in par value. BRWA investors receive periodic interest payments as follows:

Annual interest expense = Bond par value × Coupon rate.

Annual interest expense = USD300,000,000 × 3.2%.

Annual interest expense = USD9,600,000.

Semiannual interest expense = USD9,600,000/2 = USD4,800,000.

million.

Yield Measures

Given a bond's expected cash flows and its price, return or yield measures can be calculated. One simple measure is the **current yield** (CY), equal to the bond's annual coupon divided by the bond's price and expressed as a percentage. For example, if the five-year BRWA bond were trading at a price of USD101 per USD100 in face value at time *t*, its current yield would be

$$CY_t = \text{Annual coupon}_t/\text{Bond price}_t = 3.2\%/1.01 = 3.168\%.$$

The current yield is analogous to the dividend yield for an equity security.

A more complex but far more common yield measure is the **yield-to-maturity** (YTM), which is the internal rate of return (IRR) calculated using the bond's price and its expected cash flows to maturity. YTM is usually quoted as an annual rate. An investor's rate of return on a bond will equal the bond's YTM at the time of purchase as long as the investor

1. receives all promised interest and principal payments as scheduled (i.e., no default),
2. holds the bond until maturity, and
3. reinvests all periodic cash flows at the YTM.

Notice that these are the same IRR assumptions discussed in earlier modules. If any of these assumptions do not hold, the investor's rate of return on the bond investment will differ from the YTM. If the five-year BRWA bond were trading at a price of USD101 per USD100 in face value immediately after issuance, its YTM is the rate, *r*, in the following equation (recall that the 3.2% coupon is paid semiannually, or 1.6 in each of 10 periods per USD100 in face value):

$$101 = 1.6/(1 + r)^1 + 1.6/(1 + r)^2 + \ldots + 101.6/(1 + r)^{10}.$$

r = 1.49% on a semiannual basis, or 1.49% × 2 = 2.98% annualized.

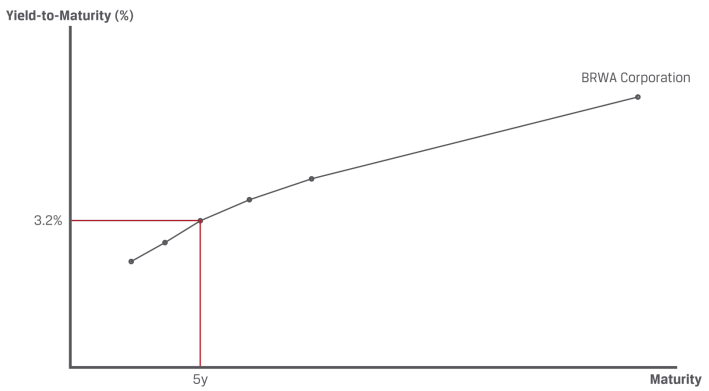
The YTM calculation is covered in depth in later modules.

Yield Curves

Most fixed-income issuers have many debt instruments outstanding. A useful way of evaluating the YTM on one issue is to graph all an issuer's debt instruments with identical features by their YTM and times to maturity. This graphical depiction results in a **yield curve**. BRWA has six bond issues outstanding, shown in Exhibit 4. The 3.2%, five-year bond is the third point from the left.

Exhibit 4:

BRWA Corporation Bond Yield Curve



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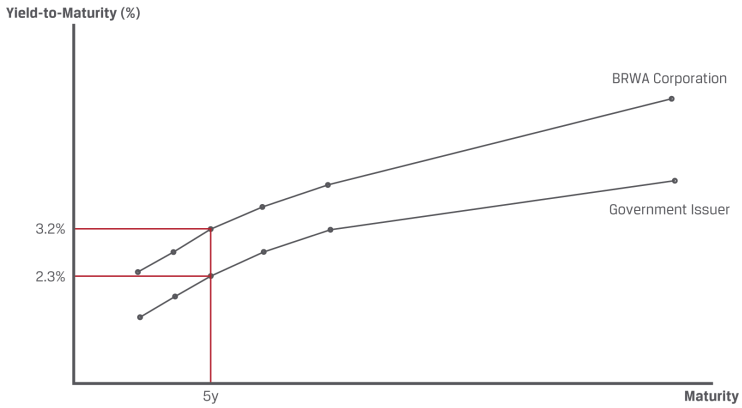
with longer-maturity bonds of the same issuer. A way to measure the credit risk (one type of investment risk of fixed-income securities) for a bond is to compare an issuer's yield curve to that of comparable sovereign bonds, which have little or no credit risk. Exhibit 5 shows the yield curve for BRWA bonds versus the yield curve for comparable sovereign bonds in the United States, which are US Treasuries.

Exhibit 5:

BRWA Corporation Yield Curve vs. Treasuries



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The yield-to-maturity difference between the five-year BRWA bond and the five-year US Treasury bond is 90 bps (= 3.2% − 2.3%), which reflects compensation that investors demand for taking the additional credit risk associated with the BRWA bonds.

Question Set

1. Match the fixed-income instrument type in the left column with its description in the right column.

A. Money market securities	I. Public and private bonds with a time to maturity of greater than one year
B. Capital market securities	II. Fixed-income instruments with maturities of less than one year, such as commercial paper
C. Perpetual bonds	III. Bonds that have no stated maturity date

Hide Solution

Solution:

A. II is correct. Money market securities are fixed-income instruments with maturities of less than one year, such as commercial paper.
B. I is correct. Capital market securities are public and private bonds with a time to maturity of greater than one year.
C. III is correct. Perpetual bonds have no stated maturity date.

2. Calculate the coupon payment on a 2.5% coupon bond with GBP100,000 par value and a semiannual payment frequency.

Hide Solution

Solution:

The correct answer is GBP1,250.

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GBP100,000 results in a GBP2,500 annual coupon. However, since the payment frequency is semiannual (twice a year), each coupon payment is GBP1,250.

3. Determine the appropriate FRN coupon and interest payable in the following example: A European corporation issues a EUR10 million FRN that pays quarterly interest equal to the three-month MRR plus 125 bps. If three-month MRR is –0.50%, what is the corporation's FRN coupon interest payable on the FRN for the period?

Hide Solution

Solution:

The answer is EUR18,750.

An FRN coupon comprises MRR plus the issuer-specific spread.

FRN coupon = MRR + Spread.

FRN coupon = –0.50% + 1.25%.

FRN coupon = 0.75%.

Annual interest = Principal × FRN coupon.

Annual interest = €10,000,000 × 0.75%.

Annual interest = €75,000, or €18,750 on a quarterly basis.

4. Fill in the blanks. The current yield is equal to a bond's annual _____ divided by the bond's price expressed as a percentage. If the bond's price were to fall, we would expect the current yield to ____.

Hide Solution

Solution:

The current yield is equal to a bond's annual *coupon* divided by the bond's price expressed as a percentage. If the bond's price were to fall, we would expect the current yield to *rise*.

5. Fill in the blanks. A downward-sloping yield curve indicates that yields-to-maturity on an issuer's longer-term bonds are _____ than yields-to-maturity on that issuer's shorter-term bonds.

Hide Solution

Solution:

A downward-sloping yield curve indicates that yields-to-maturity on an issuer's longer-term bonds are *lower* than yields-to-maturity on that issuer's shorter-term bonds.

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Discussion

What do you want to discuss... (required)

NA

Question Set 3

If quarterly interest are equal to -0.50%+1.25%. Should we multiply the result by 4 to get the annual interest amount?

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